
LECTURE NOTES

Lecture 2b - NN Intro & Gradient-based Learning

AI506 — Advanced Machine Learning

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Activation function for NN

$$\sigma(X^T W + b) \implies \sigma\left(\left(\sum x_i w_i\right) + b\right)$$

Each transformation from one layer to the next requires a vector of weights.

Gradient-based learning

Update the model parameters following the steepest slope of the loss function. We use gradient descent:

$$f(x + \varepsilon) \approx f(x) + \varepsilon \nabla f(x)$$

So with a loss function:

$$\frac{1}{N} \sum_X \nabla_{\theta} J(\theta, x_i, y_i) = \nabla_{\theta} J(\theta, X, y)$$

then:

$$\theta_{\text{new}} = \theta_{\text{old}} - \varepsilon \nabla_{\theta} J(\theta, X, y)$$